

VMAF-Aware Runtime Shielding for Constrained Deep RL in Adaptive Video Streaming

Saeed Zarbi^a, Leili Farzinvash^a, Pedram Salehpour^{a,*}

^a*Department of Computer Engineering, Faculty of Electrical and Computer Engineering,
University of Tabriz, Tabriz, Iran*

Abstract

Deep reinforcement learning (DRL) policies for client-side adaptive bitrate (ABR) streaming can achieve high average quality of experience (QoE), but individual decisions may still cause substantial rebuffering under volatile 5G/mmWave throughput. Conventional runtime shields reduce this risk by decrementing the representation index until a feasibility test passes. Because adjacent representations can differ nonuniformly in Video Multimethod Assessment Fusion (VMAF), this content-blind repair may incur unnecessary perceptual loss.

We formulate ABR as a constrained Markov decision process (CMDP) and optimize it using Lagrangian proximal policy optimization (PPO) with manifest lookahead and buffer-stabilization cues. We then introduce a rule-based VMAF-aware runtime shield that projects a proposed action onto a buffer-feasible representation. When a downgrade is required, the shield selects the feasible representation that retains the highest VMAF subject to a perceptual-loss budget τ .

We evaluate a fixed policy on 80 paired broadband episodes using identical traces and random seeds. The primary engineering result is a *quality-preserving safety improvement*. Relative to legacy index-decrement projection, the VMAF-aware shield at $\sigma_{\text{soft}}=0.8$ and $\tau=8$ reduces mean rebuffering from 5.67% to 4.98%. Mean session QoE increases by approximately 2.6%, without an observed reduction in perceptual quality. Mean VMAF instead increases slightly (+0.3 points), and the three unadjusted paired Wilcoxon tests yield $p < 0.05$. Across 47 evaluated configurations, outcomes form 11 tolerance-defined behavioral groups, with the VMAF-aware fallback accounting for the main observed changes in the empirical Pareto set. Replay on synthetic 5G/mmWave stress traces also yields statistically

*Corresponding author

URL: psalehpour@tabrizu.ac.ir (Pedram Salehpour)

significant QoE and rebuffering improvements, although VMAF differences are not statistically detectable. Finally, removing the shield from the evaluated policy increases rebuffering ($p < 10^{-10}$), supporting deployment of that policy with its associated shield.

Keywords: Adaptive Bitrate Streaming, Deep Reinforcement Learning, Constrained MDP, Runtime Shielding, VMAF, Policy–Shield Co-design, Video QoE

1. Introduction

Client-side adaptive bitrate (ABR) policies trained with deep reinforcement learning (DRL) can achieve high average surrogate rewards, but aggressive representation choices may still produce severe rebuffering on individual episodes [1, 2]. Sub-6 GHz and 5G/mmWave links can exhibit abrupt throughput reductions caused by blockage, line-of-sight/non-line-of-sight transitions, scheduling, and mobility [3, 4]. Such reductions can produce prolonged stalls that substantially affect quality of experience (QoE), even when session-level averages remain acceptable [5, 6].

This limitation is architectural rather than specific to a particular proximal policy optimization (PPO) implementation. PPO optimizes an expected objective over the training distribution, and a Lagrangian extension targets constraint satisfaction in expectation over sampled rollouts [7–9]. Neither mechanism provides a per-decision feasibility guarantee when the deployed client encounters a throughput transition that is rare or absent during training. Increasing model capacity or modifying reward weights can improve average robustness, but it does not remove this distinction between distributional optimization and decision-time enforcement [10].

One way to limit decision-time risk is to place a *runtime shield* between the learned policy and the streaming client [11]. The shield projects each proposed action before the corresponding chunk is downloaded. A common repair rule decrements the representation index until a buffer-feasibility test passes. This rule does not account for content-dependent perceptual spacing: adjacent ladder representations may differ by only a few Video Multimethod Assessment Fusion (VMAF) points or by more than ten [12]. Consequently, an index-based downgrade can incur greater perceptual loss than another feasible action available in the same manifest.

We address this limitation by treating the shield as a constrained perceptual

optimization layer. In this architecture, PPO and the shield have distinct responsibilities: PPO learns high-QoE behavior over the operating distribution, whereas the shield enforces an explicit feasibility predicate for each proposed action. The shield is therefore not a temporary correction for an inadequately trained policy; it is the decision-time enforcement layer of the controller. This separation is particularly relevant to real deployments, where the network can move outside the empirical support of the training traces and a policy confidence score does not constitute a safety certificate. The proposed VMAF-aware runtime shield searches the representation ladder whenever a downgrade is required and selects the buffer-feasible action with the highest retained VMAF. Its behavior is controlled by a soft feasibility tolerance σ_{soft} , a perceptual-loss budget τ , and a catastrophic-download threshold ρ_{cat} (Sec. 4.4). The projection is defined entirely by the buffer state, manifest chunk sizes, conservative throughput estimates, and per-chunk VMAF values. These observable inputs make each intervention auditable [11, 12].

In paired trace-driven evaluation, the VMAF-aware configurations empirically improve the reported QoE, rebuffering, and VMAF means relative to legacy index-decrement projection. At $\sigma_{\text{soft}}=0.8$ and $\tau=8$, mean rebuffering decreases from 5.67% to 4.98%, while mean QoE increases by approximately 2.6%. Mean VMAF is preserved and increases slightly by 0.3 points. All three unadjusted paired tests yield $p < 0.05$ (Sec. 5). Thus, the shield obtains a statistically supported reduction in stalls without exchanging perceptual quality for safety in the evaluated broadband episodes. Synthetic 5G/mmWave stress replay retains the QoE and rebuffering improvements but does not show a statistically detectable VMAF difference (Sec. 5.9.1). For the evaluated policy, removing the shield at inference substantially increases rebuffering, indicating that the policy and shield should be treated as a joint deployment configuration.

The main contributions are:

- **Two-layer constrained control.** We formulate ABR as a constrained Markov decision process (CMDP) and optimize expectation-level objectives using primal–dual Lagrangian PPO. A separate shield enforces configured feasibility tests at decision time, including under out-of-distribution throughput changes [7, 8, 11, 13].
- **VMAF-aware runtime shielding.** We design an auditable projection that uses the nonlinear perceptual ladder to select emergency downgrades. The parameters σ_{soft} , τ , and ρ_{cat} can be adjusted without retraining (Sec. 4.4).
- **Paired statistical evaluation.** We evaluate shield variants on $n=80$ paired

episodes. At $\sigma_{\text{soft}}=0.8$ and $\tau=8$, rebuffering falls from 5.67% to 4.98%, QoE increases, and VMAF does not decrease. All three paired differences are statistically significant, although the positive VMAF effect is small. The QoE and rebuffering results persist under synthetic 5G/mmWave stress replay (Secs. 5–5.9.1).

- **Policy–shield co-design.** We evaluate the policy and shield as complementary layers rather than interchangeable safety mechanisms. We quantify the policy’s dependence on inference-time shielding and show that 47 configurations form 11 tolerance-defined behavioral groups. The largest observed differences are associated with enabling the VMAF-aware fallback.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the system model and CMDP, and Section 4 describes Lagrangian PPO and the runtime shields. Section 5 presents the evaluation, Section 6 discusses limitations, and Section 7 concludes the paper.

2. Related work

We review three areas relevant to client-side representation selection: model-based ABR, learned ABR, and constrained control with runtime shielding. Our focus is the control policy and its runtime action projection, rather than throughput prediction. Predictors such as CS2P [14] can provide inputs to model-based or learned controllers [1, 15]. ABR surveys [5, 16] and 5G/mmWave measurements [3] further motivate control mechanisms that remain effective under volatile throughput.

2.1. Heuristic and model-based ABR

MPEG Dynamic Adaptive Streaming over HTTP (DASH) decouples packaging from transport; clients run local policies informed by ladder metadata and download observations [17]. Buffer-threshold heuristics such as BBA [18] are simple and transparent, whereas BOLA [19] derives its decisions from a Lyapunov-based utility formulation. RobustMPC [15] selects representations by solving a short-horizon model predictive control (MPC) problem under estimated future throughput. These approaches expose interpretable thresholds and horizons, but their performance can depend on manual calibration and prediction accuracy across networks and devices [3, 4]. Moreover, conventional feasibility rules generally do not use the nonlinear perceptual spacing of the encoding ladder [12]. SODA [20] addresses smoothness-aware online control, while edge assistance [21] and Oboe-style automatic tuning [22] modify complementary parts of the delivery system.

2.2. *Deep reinforcement learning for ABR*

Pensieve [1] established a trace-driven DRL formulation for ABR. Related methods include D-DASH [23] and more recent PPO-based controllers [24]. Their neural policies combine throughput histories, buffer state, and manifest features [2]. Comyco [25] uses imitation learning with a VMAF-based QoE objective, demonstrating the value of perceptual information within the learned policy. Loki [26] combines learned features with a rule-based model to reduce long-tail failures in production. Fugu [6] evaluates learned control in live networks and illustrates the importance of distribution shift between simulation and deployment. These systems motivate complementary constrained-learning and decision-time mechanisms because an expected-return objective alone does not impose an explicit per-episode stall or smoothness budget [10, 27]. Even when a learned controller is trained with explicit CMDP constraints, its guarantees and empirical performance are defined with respect to the sampled trajectory distribution. A previously unseen blockage pattern can therefore induce an unsafe proposal at an individual chunk boundary. This observation motivates a separation between a learned performance layer and a runtime enforcement layer, rather than treating additional policy training as a substitute for runtime validation.

2.3. *Safe/constrained RL, shields, and the content-blindness gap*

CMDPs and Lagrangian primal–dual methods provide a framework for optimizing expected return subject to explicit constraints [8, 9, 13]. Safe-RL surveys discuss related shield, Lyapunov, and risk-sensitive approaches [10, 28]. COREL [27] is closest to our training formulation. It models ABR over commercial 5G/mmWave traces as a CMDP with a stall-time constraint and applies a primal–dual actor–critic method. Its constraint mechanism is embedded in training rather than implemented as a separate runtime projection with intervention telemetry. Runtime shielding instead repairs a proposed action before execution and can record when such repair occurs [11].

Our method combines these two levels of control. Lagrangian PPO addresses expectation-level stall and smoothness budgets and learns how to trade fidelity, rebuffering, and switching over the training distribution. The VMAF-aware shield addresses a different requirement. Before every download, it checks the proposed action against an explicit buffer-feasibility model using current client and manifest observations. Thus, the shield does not replace learning, and learning does not replace the shield. The former provides auditable decision-time enforcement, while the latter supplies the adaptive performance needed across the broader operating regime. The distinction from perceptually aware policies such as Comyco is

that VMAF also enters the decision-time projection, rather than only the learned objective. Table 1 summarizes this positioning.

Table 1: Positioning of client-side ABR methods by constraint handling, deterministic prefetch projection, and VMAF-aware emergency logic. The symbol $\approx$ denotes partial support.

Work	Year	Paradigm	Constr.	Det. guard	VMAF
BBA [18]	2014	Heuristic	N	N	N
BOLA [19]	2018	Lyapunov util.	Imp.	N	N
RobustMPC [15]	2015	MPC / lookahead	N	N	N
Pensieve [1]	2017	DRL	N	N	N
D-DASH [23]	2017	DRL	N	N	N
Oboe [22]	2018	Meta-tuning	N	N	N
Comyco [25]	2019	Quality-aware IL	N	N	$\approx$
Fugu / in-situ [6]	2020	Field DRL	N	N	N
Loki [26]	2021	Rule-RL fusion	N	$\approx$	N
COREL [27]	2023	Constr. DRL	Y	N	N
SODA [20]	2024	Online control	N	N	N
Ours	—	CMDP + shield	Y	Y	Y

Constr.: explicit stall/smoothness budget during training (Y), implicit handling (Imp.), or no explicit budget (N). **Det. guard:** auditable action projection before chunk download [11].

VMAF: emergency logic uses per-chunk VMAF ladder spacing [12]. The symbol $\approx$ denotes partial support within a learned or fused component rather than a separate deterministic guard.

3. System model and problem formulation

3.1. Streaming architecture and control loop

We adopt the standard Hypertext Transfer Protocol (HTTP) adaptive-streaming abstraction [1, 15, 17]. A video origin or content delivery network exposes L encoded representations. We denote their indices by $\mathcal{B} = \{0, \dots, L-1\}$; the DASH/HTTP Live Streaming (HLS) manifest provides the corresponding per-chunk sizes and perceptual metadata. At decision step k , the client requests chunk k while maintaining a playout buffer B_k . The effective trace throughput $\hat{C}_k$ is exogenous to the controller. On sub-6 GHz and 5G/mmWave access, this throughput can change abruptly because of blockage, line-of-sight transitions, scheduling, and mobility [3, 4].

A neural policy π_θ maps an augmented state s_k to a stochastic proposed action $a_k^{\text{raw}} \in \mathcal{B}$. Actions are sampled during training, whereas evaluation uses the most probable action. Before chunk k is downloaded, the VMAF-aware runtime shield

$\mathcal{S}$ applies the deterministic projection

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

yielding the executed representation index a_k^{exec} and an audit bit ι_k that records whether the projection altered the proposal. Only a_k^{exec} enters the chunk download and buffer dynamics (4); Figure 1 highlights the $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path. Decisions evolve on the chunk clock with playback duration T_c per step (e.g., 4 s in our harness) [1, 15].

Lagrangian PPO addresses stall and smoothness budgets in expectation over rollouts by adapting dual variables λ_r and λ_s . It does not provide a per-step feasibility guarantee, particularly under throughput conditions outside the training distribution [3, 8, 9]. The shield provides a complementary deterministic repair: for each proposal, it selects an action that satisfies explicit buffer-margin tests [11].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . The observation includes buffer occupancy, past throughput samples, and the previous decision. It also includes manifest lookahead: upcoming chunk sizes across the representation ladder and spatial/temporal complexity tags (SI/TI) [1, 15, 17]. The state also includes the per-chunk VMAF ladder $\{V_k(j)\}_{j \in \mathcal{B}}$, where $V_k(j)$ is the perceptual score for chunk k at representation j [12]. Without $\{V_k(j)\}$, neither the policy nor the shield could exploit content-dependent ladder spacing for emergency downgrade decisions.

Nonlinear, content-dependent fidelity. For each chunk k , the map $j \mapsto V_k(j)$ is not affine in the discrete index j . It also varies across chunks because different group-of-pictures (GOP) and texture structures produce different marginal VMAF gains between adjacent rungs. The mapping from nominal bitrate to VMAF is therefore nonlinear and content dependent. Consequently, legacy index-decrement projection can retain less perceptual quality than a VMAF-aware search over the same feasible set [12].

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed representation index for chunk k ; the environment executes only a_k^{exec} .

QoE surrogate. Following RobustMPC/Pensieve reporting conventions [1, 15], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \alpha \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ is the absolute VMAF change between adjacent chunks and τ_k^{stall} is the stall duration at step k . For the first chunk, we set the smoothness

term to zero. The shared evaluation harness fixes $\beta=4.3$ per stall second and $\alpha=1.0$ per absolute VMAF-point change. The rebuffering weight follows a Pensieve-style penalty rescaled to the VMAF fidelity axis. Session QoE is $\sum_{k \in \mathcal{K}_e} \text{QoE}_k^{(\text{rep})}$ (Section 5). The CMDP objective below separates the fidelity objective $\sum q_k$ from explicit constraints on rebuffering and smoothness [8, 13].

3.3. Chunk-timescale dynamics

3.3.1. Throughput and latency

Let $\hat{C}_k > 0$ denote the trace-derived effective throughput, in kb/s, for chunk k . All methods receive the same throughput trace in paired evaluations. Choosing a_k^{exec} determines the manifest payload size $\sigma_k(a_k^{\text{exec}})$ in bits. With $\tilde{C}_k = \max\{\hat{C}_k, \epsilon\}$, $\epsilon > 0$, the nominal download duration is $d_k = \sigma_k(a_k^{\text{exec}})/(10^3 \tilde{C}_k)$ seconds [15].

3.3.2. Buffer and stall

Let $B_k \geq 0$ denote the buffered playback time at the start of the download of chunk k [1, 15]. Rebuffering (stall) and buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Randomness across episodes arises from trace and initial-condition sampling [16, 29, 30]. Conditional on $(s_k, a_k^{\text{exec}}, \hat{C}_k)$, each transition is deterministic.

3.4. CMDP costs, safety budgets, and Lagrangian relaxation

An unconstrained MDP maximizes $\mathbb{E}[\sum_k \gamma^k r_k]$ alone. We instead model streaming as a constrained MDP (CMDP) that maximizes expected fidelity subject to budgets on trajectory-level auxiliary costs [13]. Let $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ denote the instantaneous stall cost (seconds of stall on step k) and $c_k^{\text{sm}} \equiv \text{smooth}_k$ the smoothness cost. A canonical discounted formulation uses discount $\gamma \in (0, 1]$, trajectory distribution $\mathcal{D}$, and budgets d_{rf} and d_{sm} :

$$\begin{aligned} \mathbb{E}_{\pi_\theta, \mathcal{D}} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{stall}} \right] &\leq d_{\text{rf}}, \\ \mathbb{E}_{\pi_\theta, \mathcal{D}} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{sm}} \right] &\leq d_{\text{sm}}. \end{aligned} \quad (5)$$

For logging and dual updates, we use the undiscounted episodic rebuffering ratio $\text{RB}(e)$ and mean smoothness cost $\text{SM}(e)$ over $\mathcal{K}_e = \{1, \dots, K_e\}$ [1, 8]. Let $T_e^{\text{play}} =$

$K_e T_c$. Then

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}. \quad (6)$$

The constrained program motivating training is

$$\max_{\pi_\theta} \mathbb{E}_{\pi_\theta, \mathcal{D}} \left[\sum_{k \in \mathcal{K}_e} q_k \right], \quad (7)$$

$$\text{s.t. } \mathbb{E}_{\pi_\theta, \mathcal{D}} [\text{RB}(e)] \leq \delta_{\text{rf}}, \quad (8)$$

$$\mathbb{E}_{\pi_\theta, \mathcal{D}} [\text{SM}(e)] \leq \delta_{\text{sm}}, \quad (9)$$

with targets fixed at $\delta_{\text{rf}}=0.05$ (rebuffering ratio) and $\delta_{\text{sm}}=3.5$ (mean absolute VMAF change per chunk) [15].

Because these constraints are generally difficult to enforce directly during nonconvex policy optimization, we form a Lagrangian relaxation. We define the per-step normalized costs $\widetilde{\text{RB}}_k = c_k^{\text{stall}}/T_e^{\text{play}}$ and $\widetilde{\text{SM}}_k = c_k^{\text{sm}}/K_e$. Their episodic sums equal $\text{RB}(e)$ and $\text{SM}(e)$, respectively:

$$\mathcal{L}(\theta, \lambda_r, \lambda_s) = \mathbb{E}_{\pi_\theta, \mathcal{D}} \left[\sum_{k \in \mathcal{K}_e} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k) \right]. \quad (10)$$

where $\lambda_r, \lambda_s \geq 0$ are adapted by projected dual ascent on empirical constraint gaps $g_r = \widetilde{\text{RB}} - \delta_{\text{rf}}$, $g_s = \widetilde{\text{SM}} - \delta_{\text{sm}}$ [8, 13]. The PPO reward is defined as $q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k$, with optional stabilizers described in Section 4. Finite-sample PPO with clipped objectives does not certify constraint satisfaction at every iterate [7, 9]; this gap motivates the hard shield in (1).

Remark (auxiliary Lyapunov cues). Training optionally subtracts buffer-deviation and simulator Lyapunov penalties [31] outside (λ_r, λ_s) ; these serve as stabilizers, not formal certificates.

3.5. VMAF-aware shield as a hard projection

Let $\mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) \subseteq \mathcal{B}$ denote the representation indices that do not exceed a_k^{raw} and pass the shield’s buffer-margin tests (Sec. 4.4). The set is computed from pessimistic throughput, manifest sizes $\sigma_k(\cdot)$, and buffer B_k . Legacy index-decrement projection repeatedly decreases a_k^{raw} until it finds a feasible index [11].

VMAF-aware projection. Among feasible rungs, we select the one that maximizes perceptual score subject to a VMAF-loss budget τ relative to the proposal:

$$a_k^{\text{exec}} = \arg \max_{\substack{j \in \mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) \\ V_k(a_k^{\text{raw}}) - V_k(j) \leq \tau}} V_k(j). \quad (11)$$

τ has a direct perceptual interpretation: it is the maximum VMAF-point reduction admitted relative to the policy proposal before the fallback rule is invoked. It therefore controls how much perceptual quality the shield may exchange for additional buffer protection. If no safe candidate satisfies the VMAF-loss budget, the shield selects the lexicographically best candidate from $\mathcal{A}_{\text{safe},k}$; if the safe set itself is empty, it selects index 0 [12]. Thus, $\mathcal{S}$ is a deterministic perceptual projection that maximizes retained VMAF over the available safe candidates rather than selecting solely by index order. Section 4.4 specifies the catastrophic-download threshold ρ_{cat} , throughput pessimism factor ρ , and buffer margins.

4. Method: Lagrangian PPO and runtime shields

We implement the controller using Stable-Baselines 3 (SB3) PPO and a custom Lagrangian reward wrapper [7, 32]. The implementation includes two deterministic shield variants: a QoE-oriented core (Algorithm 1) and a VMAF-aware projection (Algorithm 2). Unless stated otherwise, symbols follow Section 3.

4.1. Observation interface and policy outputs

The actor–critic receives the observation tensors shown in Figure 1: throughput history, buffer level, previous action, and manifest lookahead containing SI/TI tags and per-chunk VMAF ladders [1, 12, 15, 17]. The actor outputs a categorical distribution over raw proposals $a_k^{\text{raw}} \in \mathcal{B}$. Optional hysteresis or shield-aware shaping wrappers may alter sampling incentives, but the executed action is always $a_k^{\text{exec}} = \mathcal{S}(s_k, a_k^{\text{raw}})$ when shielding is enabled [11].

4.2. Lagrangian PPO agent with Lyapunov-style buffer regularization

PPO backbone. We use clipped PPO with separate policy and value multilayer perceptrons (MLPs) using Tanh activations [7]. Optimization uses generalized advantage estimation, surrogate clipping, entropy regularization, and an Adam learning-rate schedule (Table 2) [32].

CMDP shaping and stability cues. The train-time signal is the Lagrangian-relaxed objective (10) instantiated per step by (12). In addition to the dual-weighted

rebuffering and smoothness costs, training uses a Lyapunov-style buffer regularizer $\phi(B_k)$ and an emulator-derived drift cue ψ_k^{Lya} , with fixed coefficients w_b and w_ℓ [28, 31]. The buffer term penalizes deviations toward low-buffer states during early exploration, while the drift cue provides a queue-stability signal from the emulator. These auxiliary terms are not safety certificates; they are intended to stabilize optimization while the dual variables adapt [8, 10].

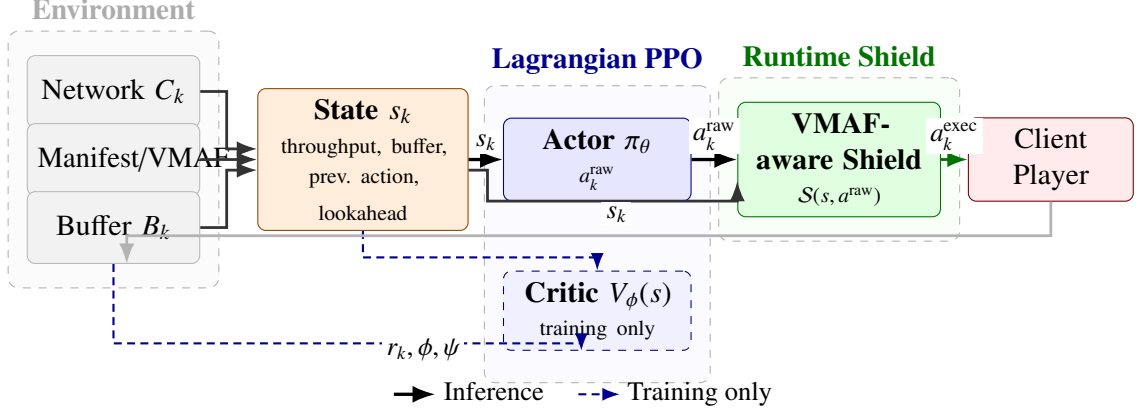


Figure 1: End-to-end control loop. Solid arrows mark the deployed inference path $s_k \rightarrow \pi_\theta \rightarrow a_k^{\text{raw}} \rightarrow S(s_k, a_k^{\text{raw}}) \rightarrow a_k^{\text{exec}}$; the shield consumes both the proposal and the augmented state (buffer, throughput, VMAF ladder). Dashed arrows mark training-only signals (critic, reward r_k , and Lagrangian cues ϕ, ψ^{Lya}).

4.2.1. Lagrangian reward channel and dual ascent

At each step, the environment records q_k , stall duration, smoothness cost, $\phi(B_k)$, and ψ_k^{Lya} . The normalized stall and smoothness values correspond to $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ in (10). The PPO surrogate reward is (scaled by 1/100 for SB3 numerical stability)

$$r_k^{\text{train}} = \frac{1}{100} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}}). \quad (12)$$

After a 60-episode warm-up, the episodic estimates $\hat{\text{RB}}$ and $\hat{\text{SM}}$ determine projected dual-ascent updates from the gaps $g_r = \hat{\text{RB}} - \delta_{\text{rf}}$ and $g_s = \hat{\text{SM}} - \delta_{\text{sm}}$. The dual step sizes are $\eta_r=0.012$ and $\eta_s=0.003$, with initial values $\lambda_r=6.0$ and $\lambda_s=0.7$. We project the variables onto $\lambda_r \in [4.3, 40]$ and $\lambda_s \in [0.3, 2.5]$; the lower bound on λ_r matches the evaluation-harness stall weight β [8, 10]. All vectorized workers share the current dual variables.

4.3. Catastrophic risk detection and pessimistic feasibility

The shields have no learnable parameters. Each evaluates deterministic predicates using manifest data, buffer state, and a conservative throughput estimate. Let $\overline{\text{tp}}_k$ and last_tp_k denote the current and most recently observed trace throughput, respectively, in kb/s. The pessimistic estimate is

$$\underline{C}_k = \max\{C_{\min}, \rho \cdot \min\{\overline{\text{tp}}_k, \text{last_tp}_k\}\}, \quad (13)$$

with pessimism $\rho \in (0, 1]$ and an emulator throughput floor $C_{\min}$. Default values are $\rho=0.97$ for training and $\rho=0.90$ for the VMAF-aware sweep reported in Section 5. For candidate representation j , predicted download seconds are

$$d_k^{\text{est}}(j) = \min\left\{\frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3}, 60\right\}, \quad (14)$$

where $\sigma_k(j)$ is the manifest chunk size in bits for representation j . The corresponding post-download buffer estimate is

$$\widetilde{B}_{k+1}(j) = \max\{0, B_k - d_k^{\text{est}}(j)\} + T_c. \quad (15)$$

Representation j is deemed infeasible for emergency repair when $\widetilde{B}_{k+1}(j) < B_{\text{safe}}$.

Risk gate. When enabled, the shield passes through a_k^{raw} unchanged whenever

$$d_k^{\text{est}}(a_k^{\text{raw}}) \leq \max\{B_k, 0.1\} \cdot \rho_{\text{risk}}, \quad (16)$$

with $\rho_{\text{risk}}=1.35$ as default; benign proposals therefore bypass projection [11].

Catastrophic-download threshold. In the VMAF-aware path, if $d_k^{\text{est}}(a_k^{\text{raw}}) \leq B_k \rho_{\text{cat}}$, the shield accepts a_k^{raw} without constructing the VMAF candidate pool (Algorithm 2). The `light` mode uses the same inequality as a fast no-intervention check before margin-based repair.

Critical buffer. If $B_k \leq B_{\text{crit}}$ (configured between 0.20 and 0.30 s), the shield selects index 0.

4.4. Deterministic runtime projection

Section 3 defines the shield map $\mathcal{S}$ and the VMAF-aware selection rule (11). This subsection gives executable forms of the predicates and their default parameters. The indicator $\iota_k \in \{0, 1\}$ records each intervention for subsequent analysis (Section 5).

4.4.1. QoE-oriented shield core

Besides off, the shield exposes three operating modes: qoe, light, and strong. All modes use $\underline{C}_k$ and $d_k^{\text{est}}(\cdot)$ from (13)–(14). They clip a_k^{raw} to $\mathcal{B}$, apply the critical-buffer rule, and honor the optional risk gate (16). The qoe mode searches downward from a_k^{raw} to $j_{\min}$, the lowest non-emergency representation allowed by the wrapper. The parameter ζ is the admissible predicted-stall slack: candidate c is feasible when $d_k^{\text{est}}(c) \leq B_k + \zeta$, with default $\zeta = 0.25$ s. The parameter ξ is the residual-stall threshold for the hard emergency fallback. If $e_{\text{emg}} = 1$ and $\max\{0, d_k^{\text{est}}(j_{\min}) - B_k\} > \xi$, the shield selects index 0; the default is $\xi = 1.25$ s. Within the QoE-oriented core, the light/strong modes enforce buffer-margin tests. Their defaults are $\rho_{\text{cat}} = 2.50$, $\mu_\ell = 0.15$ s, and $\mu_s = 0.75$ s. The VMAF-aware sweep uses the separate grid specified below. Algorithm 1 summarizes the procedure (intervention bit $\iota = 1$ iff the executed index differs from the proposal); the wrapper may additionally cap post-intervention upshifts during recovery.

4.4.2. VMAF-aware perceptual projection

Algorithm 2 implements (11) with three engineering details beyond the abstract feasible set $\mathcal{A}_{\text{safe},k}$. First, the soft-safe candidate set never upswitches and permits a multiplicative tolerance σ_{soft} in the buffer test:

$$\mathcal{J}_k = \left\{ j \in \mathcal{B} : j \leq a_k^{\text{raw}}, d_k^{\text{est}}(j) \leq B_k \sigma_{\text{soft}} \right\}. \quad (17)$$

This set instantiates $\mathcal{A}_{\text{safe},k}$. The parameter σ_{soft} controls buffer conservatism: $\sigma_{\text{soft}} = 1$ requires the predicted download to fit within the current buffer, whereas $\sigma_{\text{soft}} = 0.8$ reserves approximately 20% of the buffer under the pessimistic download-time estimate. Values above one admit a limited predicted buffer overrun to retain perceptual quality. When a VMAF entry is unavailable, the implementation uses a monotone proxy derived from the representation index. Second, candidates that satisfy the VMAF-loss budget are compared lexicographically by $(V_k(j), j, -d_k^{\text{est}}(j))$: higher VMAF is preferred, followed by a higher index and a shorter predicted download. If $\mathcal{J}_k$ is empty, the shield selects index 0. Third, an optional H -step pessimistic lookahead gate can bypass projection before the VMAF branch if the buffer never dips below a configured floor.

Physical role of ρ_{cat} . The ratio $d_k^{\text{est}}(a)/B_k$ compares predicted download duration with available playback time. The VMAF branch is activated only when this ratio exceeds ρ_{cat} ; increasing ρ_{cat} therefore makes the shield less intervention-prone, whereas a lower value reacts earlier to predicted buffer depletion.

Parameter-selection protocol. The reported values are obtained from a finite sensitivity grid rather than learned jointly with PPO. Section 5.7 evaluates 47 con-

Algorithm 1 QoE-oriented shield core (compact)

```
1: Input:  $a^{\text{raw}}, B_k, \sigma_k(\cdot)$ , operating mode, emergency-enable flag  $e_{\text{emg}}$ 
2: Parameters:  $j_{\min}, B_{\text{crit}}, \zeta, \xi, \rho_{\text{cat}}, \mu_\ell, \mu_s$ 
3: Output:  $(a^{\text{exec}}, \iota)$  with  $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$ 
4:  $a \leftarrow \text{clip}(a^{\text{raw}})$ ; compute  $\underline{C}_k, d_k^{\text{est}}(\cdot)$  via (13)–(14)
5: if mode= off then
6:   return  $(a, 0)$ 
7: end if
8: if  $B_k \leq B_{\text{crit}}$  then
9:   return  $(0, \mathbf{1}[a \neq 0])$ 
10: end if
11: if risk gate (16) passes then
12:   return  $(a, 0)$ 
13: end if
14: if mode= qoe then
15:   for  $c = a, a-1, \dots, j_{\min}$  do
16:     if  $d_k^{\text{est}}(c) \leq B_k + \zeta$  then
17:       return  $(c, \mathbf{1}[c \neq a])$ 
18:     end if
19:   end for
20:   if  $e_{\text{emg}}$  and  $\max\{0, d_k^{\text{est}}(j_{\min}) - B_k\} > \xi$  then
21:     return  $(0, \mathbf{1}[a \neq 0])$ 
22:   end if
23:   return  $(j_{\min}, \mathbf{1}[a \neq j_{\min}])$ 
24: end if
25: if mode= light then
26:   if  $d_k^{\text{est}}(a) \leq B_k \rho_{\text{cat}}$  then
27:     return  $(a, 0)$ 
28:   end if
29:   for  $c = a-1, \dots, j_{\min}$  do
30:     if  $d_k^{\text{est}}(c) \leq \max\{B_k - \mu_\ell, 0\}$  then
31:       return  $(c, 1)$ 
32:     end if
33:   end for
34:   return  $(j_{\min}, \mathbf{1}[a \neq j_{\min}])$ 
35: end if
36:  $c \leftarrow a$  {strong mode}
37: while  $c > j_{\min}$  and  $d_k^{\text{est}}(c) > \max\{B_k - \mu_s, 0\}$  do
38:    $c \leftarrow c-1$ 
39: end while
40: return  $(c, \mathbf{1}[c \neq a])$ 
```

figurations on identical paired episodes. The grid includes $\sigma_{\text{soft}} \in \{0.8, 1.0, 1.2, 1.5\}$, $\tau \in \{4, 8, 12\}$, and $\rho_{\text{cat}} \in \{2, 3, 4, 5\}$. The VMAF budget $\tau=8$ is the middle tested setting: $\tau=4$ imposes a stricter perceptual-loss constraint and can leave no candi-

Algorithm 2 VMAF-aware perceptual projection (compact)

```
1: Input:  $a^{\text{raw}}, B_k, \sigma_k(\cdot), \{V_k(j)\}$ ; parameters  $(\rho, \rho_{\text{cat}}, \sigma_{\text{soft}}, \tau, B_{\text{crit}})$ 
2: Output:  $(a^{\text{exec}}, \iota)$  with  $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$ 
3:  $a \leftarrow \text{clip}(a^{\text{raw}})$ ; compute  $\underline{C}_k, d_k^{\text{est}}(\cdot)$  via (13)–(14)
4: if  $B_k \leq B_{\text{crit}}$  then
5:   return  $(0, \mathbf{1}[a \neq 0])$ 
6: end if
7: if risk gate (16) passes then
8:   return  $(a, 0)$ 
9: end if
10: if VMAF awareness is disabled then
11:   apply Alg. 1 in light/strong mode; return
12: end if
13: if  $d_k^{\text{est}}(a) \leq B_k \rho_{\text{cat}}$  then
14:   return  $(a, 0)$ 
15: end if
16: construct  $\mathcal{J}_k$  using (17)
17: if  $\mathcal{J}_k = \emptyset$  then
18:   return  $(0, \mathbf{1}[a \neq 0])$ 
19: end if
20:  $\mathcal{J}'_k \leftarrow \{j \in \mathcal{J}_k : V_k(a) - V_k(j) \leq \tau\}$ 
21:  $\mathcal{J}^*_k \leftarrow \mathcal{J}'_k$  if  $\mathcal{J}'_k \neq \emptyset$ ; otherwise  $\mathcal{J}^*_k \leftarrow \mathcal{J}_k$ 
22:  $a^{\text{exec}} \leftarrow \arg \max_{j \in \mathcal{J}^*_k} (V_k(j), j, -d_k^{\text{est}}(j))$ 
23: return  $(a^{\text{exec}}, \mathbf{1}[a^{\text{exec}} \neq a])$ 
```

date inside the budget, whereas $\tau=12$ is more permissive of quality reductions. We report $(\sigma_{\text{soft}}, \tau, \rho_{\text{cat}}) = (0.8, 8, 2)$ as the primary broadband operating point. It lies on the empirical nondominated set and reduces rebuffering while increasing QoE without reducing mean VMAF. This is an empirically selected point within the tested grid, not a claim of global optimality. Because the same episode suite is used to compare configurations, the selection should be interpreted as exploratory.

4.5. Optional shield-aware shaping and hysteresis

An optional shaping wrapper penalizes interventions by $\beta_{\text{sh}} \iota_k + \gamma_{\text{sh}} |a_k^{\text{exec}} - a_k^{\text{raw}}|$, where $\beta_{\text{sh}}=0.08$ and $\gamma_{\text{sh}}=0.03$. A separate hysteresis rule limits index changes, blocks upswitches when $B_k < 1.5$ s, and can restrict upshifts during post-intervention recovery.

4.6. End-to-end training loop (primal–dual PPO + optional shield)

Algorithm 3 combines these wrappers with SB3-PPO; the hyperparameters are listed in Table 2.

Algorithm 3 Lagrangian PPO with optional shielding during training

```
1: Input: SB3-PPO( $\theta$ ); Lagrangian wrapper; optional shield/hysteresis
2: for each training iteration do
3:   for each parallel worker do
4:     reset the environment; load shared dual variables ( $\lambda_r, \lambda_s$ )
5:     for  $k = 1:K_e$  do
6:       sample  $a_k^{\text{raw}} \sim \pi_\theta(\cdot | s_k)$ 
7:       set  $a_k^{\text{exec}} \leftarrow \mathcal{S}(s_k, a_k^{\text{raw}})$  if shielding is enabled; otherwise set  $a_k^{\text{exec}} \leftarrow a_k^{\text{raw}}$ 
8:       execute  $a_k^{\text{exec}}$ , compute  $r_k^{\text{train}}$  using (12), and store the transition
9:     end for
10:    compute RB and SM; after warm-up, update ( $\lambda_r, \lambda_s$ ) by projected dual ascent
11:  end for
12:  update PPO using GAE, the clipped surrogate, and entropy regularization
13: end for
```

4.7. Policy–shield co-design strategy

Lagrangian PPO targets the CMDP budgets in expectation over rollouts, but individual policy proposals can still violate the shield’s decision-time feasibility tests under throughput conditions outside the training distribution [3, 8]. The deterministic shield repairs such proposals before execution [11].

Co-design. Training with shielding changes the action sequences observed by the policy and value function. The QoE-oriented guard (Algorithm 1) pairs with Lagrangian PPO during training, while the VMAF-aware perceptual projection (Algorithm 2) constrains catastrophic downloads under explicit VMAF budgets at inference [12]. For the evaluated policy, removing the shield at inference increases rebuffering (Section 5.8). The reported operating points therefore describe the joint policy–shield system rather than either component in isolation. Deployment should retain the shield used during training and adjust σ_{soft} , τ , and ρ_{cat} instead of disabling projection.

4.8. Implementation notes and hyperparameters

The implementation consists of Lagrangian reward wrappers and deterministic shield modules. Table 2 summarizes the PPO and Lagrangian hyperparameters, including the constraint targets, dual step sizes, and evaluation-harness QoE weights; remaining shield ratios are documented alongside the code in the reproducibility repository.

Table 2: PPO, Lagrangian, and shield hyperparameters used in the reported evaluation.

Item	Setting
Policy / value MLP	Two Tanh layers, width 256 each [7]
Rollout horizon	4096 · 8 parallel environment steps per PPO iteration
Learning rate schedule	Piecewise-linear decay $3 \times 10^{-4} \rightarrow 10^{-5}$
Advantage	GAE ($\gamma=0.99$, $\lambda_{\text{GAE}}=0.95$)
Constraint targets	$\delta_{\text{rt}}=0.05$ (rebuffering ratio); $\delta_{\text{sm}}=3.5$ (mean absolute VMAF change/chunk)
Dual ascent	$\eta_r=0.012$, $\eta_s=0.003$; $\lambda_r \in [4.3, 40]$, $\lambda_s \in [0.3, 2.5]$; warm-up 60 episodes
QoE weights (harness)	$\beta=4.3$ per stall second; $\alpha=1.0$ per VMAF switch point
Auxiliary stabilizers	$w_b=0.03$ (buffer deviation); $w_\ell=0.4$ (Lyapunov drift cue)
Entropy coefficient	0.04 for the proposed policy (lower values for ablations)
Reward scaling	Divide shaped reward by 100 for SB3 numerical range
Shield sweep	$\sigma_{\text{soft}} \in \{0.8, 1.0, 1.2, 1.5\}$; $\tau \in \{4, 8, 12\}$; $\rho_{\text{cat}} \in \{2, 3, 4, 5\}$
Primary broadband point	$\sigma_{\text{soft}}=0.8$, $\tau=8$, $\rho_{\text{cat}}=2$; empirical nondominated point
Throughput pessimism ρ	0.90 (VMAF-aware sweep); 0.97 (training; Sec. 4.3)

5. Experimental evaluation

We organize the evaluation around five research questions:

- RQ1** *Pareto comparison.* How do VMAF-aware configurations compare with legacy index-decrement projection across QoE, rebuffering, and VMAF?
- RQ2** *Baseline competitiveness.* Where do the recommended operating points sit relative to RobustMPC and Pensieve?
- RQ3** *Component ablations.* How does removing manifest lookahead, Lyapunov cues, or Lagrangian dual updates change performance?
- RQ4** *Shield design-space structure.* Do the 47 evaluated configurations form a smaller number of tolerance-defined behavioral groups?
- RQ5** *Policy–shield co-design.* How does inference-time shield removal affect a policy trained with shielding?

5.1. Experimental setup and datasets

We evaluate on a multi-video trace-driven emulator ($K_e=48$ chunks per episode, fixed segment duration T_c as in Sec. 3).

Primary broadband suite. We use partitioned FCC/Norway-style throughput traces from public releases [29, 30]. Each method is evaluated on four videos and 20 paired random seeds after the warm-up period, giving 80 episodes.

Out-of-distribution stress suite. Synthetic 5G/mmWave stress traces contain intermittent line-of-sight/non-line-of-sight (LOS/NLOS) transitions and outage intervals designed to evaluate behavior under severe throughput variation [3]. Section 5.9.1 reports paired replay of the same constrained policy under the same evaluation assumptions.

Baselines. We implement Pensieve with observation masking and RobustMPC following [15]. Genie [1] uses noncausal trace knowledge and is reported only as an oracle reference.

Shield pairing protocol. Comparisons between legacy and VMAF-aware projection hold the trained policy, ladder, traces, and video–episode identifiers fixed. Only the inference-time shield logic changes, enabling paired episode-level tests.

5.2. Performance headline and baselines

Table 3 summarizes the primary comparison: classical baselines (distinct policies) versus the same constrained policy under legacy and VMAF-aware projection. The upper block contains distinct policies, whereas the lower block holds the policy fixed and varies only the shield. Cross-policy comparisons are paired by episode but are interpreted as contextual contrasts; the lower-block comparison directly isolates the shield logic. Extended results including Genie and ablated agents appear in Table 4. Figure 2 illustrates the method-level mean session QoE and rebuffering ratio, highlighting the competitive performance of the proposed CMDP policy against distinct baselines.

Table 3: Performance of distinct-policy baselines and a fixed CMDP policy under two shield configurations ($n=80$ episodes per row).

Method	QoE	VMAF	Rebuf. (%)	Interv. (%)
<i>Baselines (distinct policies; 95% CI, $n=80$)</i>				
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	—
RobustMPC [15]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	—
<i>Same CMDP policy (paired replay; CI)</i>				
+ Legacy (index decrement)	2518.87 [2176.5, 2854.7]	65.82	5.67 [4.95, 6.36]	3.23
+ VMAF-aware ($\sigma_{\text{soft}}=1.0$, $\tau=8$)	2561.00 [2216.3, 2900.5]	66.04	5.12 [4.41, 5.84]	2.71

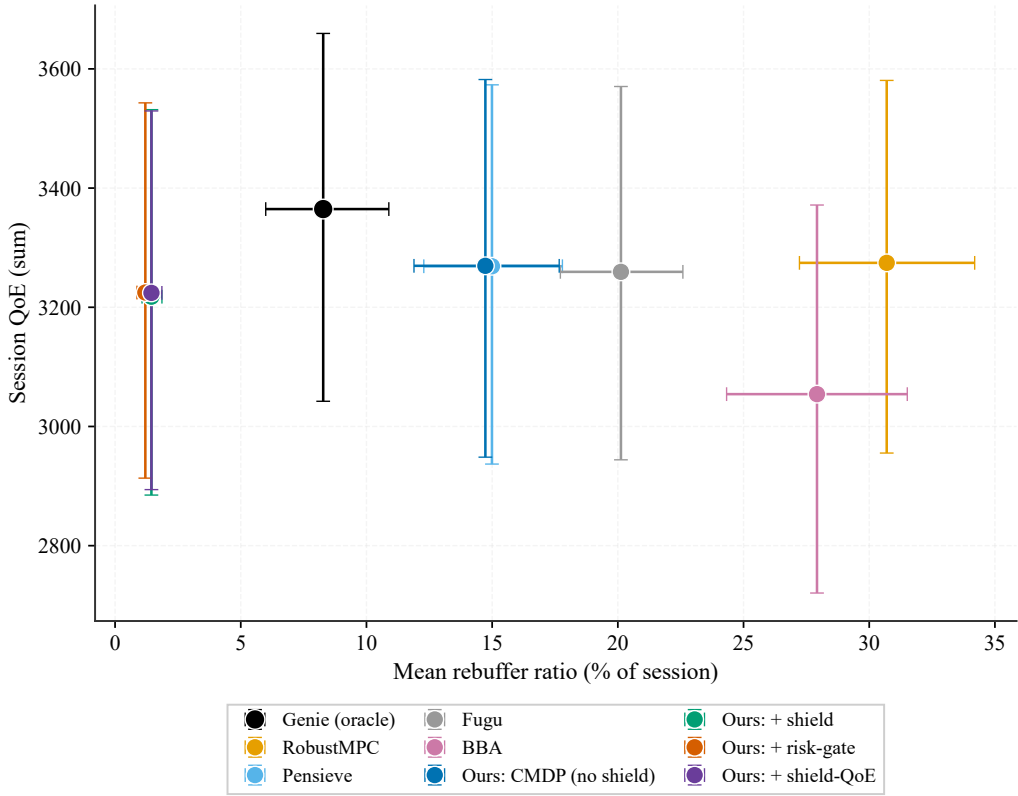


Figure 2: Method-level mean session QoE and rebuffering ratio with 95% bootstrap confidence intervals ($n=80$ episodes per method).

5.3. Overall performance (extended means)

Table 4: Extended distinct-policy results with 95% bootstrap confidence intervals ($n=80$ episodes per row).

Method	QoE $\uparrow$	VMAF $\uparrow$	Rebuf. (%) $\downarrow$	Switches $\downarrow$	Type
Genie (oracle) [1]	3365 [3043, 3663]	72.15 [65.44, 78.26]	8.28 [6.02, 10.95]	3.7 [3.2, 4.1]	oracle
RobustMPC [15]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	8.4 [7.7, 9.1]	MPC
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	6.4 [5.6, 7.2]	DRL
Fugu [6]	3259 [2924, 3558]	74.05 [67.36, 80.05]	20.12 [17.75, 22.63]	14.9 [13.7, 16.1]	IL
BBA [18]	3054 [2718, 3373]	72.89 [66.62, 79.11]	27.92 [24.35, 31.43]	25.6 [24.2, 27.0]	heur.
Ours: CMDP (no shield)	3270 [2945, 3569]	71.65 [65.06, 78.14]	14.73 [11.73, 17.78]	6.5 [5.7, 7.3]	RL+CMDP
Ours: + shield	3218 [2889, 3529]	68.47 [61.91, 75.05]	1.45 [1.06, 1.86]	5.2 [4.5, 6.0]	RL+CMDP+S
Ours: + risk-gate	3225 [2880, 3536]	68.61 [61.95, 75.07]	1.20 [0.85, 1.60]	6.3 [5.2, 7.5]	RL+CMDP+RG
Ours: + shield-QoE †	3224 [2896, 3536]	68.74 [61.87, 75.14]	1.45 [1.07, 1.87]	9.9 [8.7, 11.4]	RL+CMDP+Q

Interval: bootstrap 95% CI on episode means ($n=80$ per row). † Shield-aware variant (elevated bitrate switches).

Intervals are 95% bootstrap confidence intervals for the mean across 80 episodes. † Shield-QoE denotes the variant with elevated representation switching. Sw. = mean number of representation switches; S, RG, and Q denote shield, risk gate, and shield-QoE, respectively.

Table 5: Paired episode-level QoE comparisons with Pensieve and RobustMPC using two-sided Wilcoxon signed-rank tests.

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ QoE	p	med. Δ QoE	p
Ours: CMDP (no shield)	6.0	0.259	8.6	0.124
Ours: + shield	-59.4	3.42×10^{-10}	-55.7	1.34×10^{-9}
Ours: + risk-gate	-42.6	7.07×10^{-7}	-37.9	6.90×10^{-7}
Ours: + shield-QoE †	-35.3	4.94×10^{-8}	-40.9	7.31×10^{-6}

Two-sided paired Wilcoxon signed-rank; Δ QoE is the per-episode paired difference (method minus baseline).

Table 6: Paired episode-level rebuffering-ratio comparisons with Pensieve and RobustMPC using two-sided Wilcoxon signed-rank tests.

Method	vs. Pensieve		vs. RobustMPC	
	med. $\Delta\text{Rebuf.}$ (%)	p	med. $\Delta\text{Rebuf.}$ (%)	p
Ours: CMDP (no shield)	-0.11	0.310	-11.11	2.27×10^{-12}
Ours: + shield	-10.25	4.58×10^{-14}	-29.86	7.85×10^{-15}
Ours: + risk-gate	-10.39	4.23×10^{-14}	-29.86	7.85×10^{-15}
Ours: + shield-QoE [†]	-10.73	7.90×10^{-14}	-30.23	7.85×10^{-15}

The pairing unit is the video-seed episode. $\Delta\text{Rebuffer}$ is the method-minus-baseline difference in rebuffering ratio, measured in percentage points; negative values favor the row method.

Figure 3 reports mean session QoE by video and method. The motion-heavy crowd_run sequence yields lower QoE across controllers, while the proposed CMDP variants remain competitive with Pensieve and RobustMPC on the remaining titles.

RQ2: Cross-baseline context. Under the same evaluation harness, the VMAF-aware operating point has a mean rebuffering ratio 9.87 percentage points below Pensieve (14.99%) and 25.58 percentage points below RobustMPC (30.70%). Because these methods use distinct trained policies, the comparisons characterize end-to-end controllers rather than isolate the shield. Tables 4–6 report the paired episode-level contrasts, and Figure 4 shows the corresponding QoE differences relative to Pensieve and RobustMPC.

5.4. Statistical significance (paired shield vs. legacy)

Because means do not describe episode-level variation, we additionally compare paired outcomes using two-sided Wilcoxon signed-rank tests. Table 7 reports the results for the VMAF-aware operating point with $\sigma_{\text{soft}}=1.0$ and $\tau=8$. This configuration is denoted `tol1.0_bud08` in the evaluation artifacts and is compared with legacy projection on $n=80$ paired episodes.

Result. QoE and rebuffering improve with $p < 0.01$. The VMAF test yields $p = 0.056$ and therefore does not meet the $\alpha = 0.05$ threshold. For the stricter soft tolerance $\sigma_{\text{soft}}=0.8$, all three unadjusted tests yield $p < 0.05$ (VMAF $p = 0.041$; Table 8).

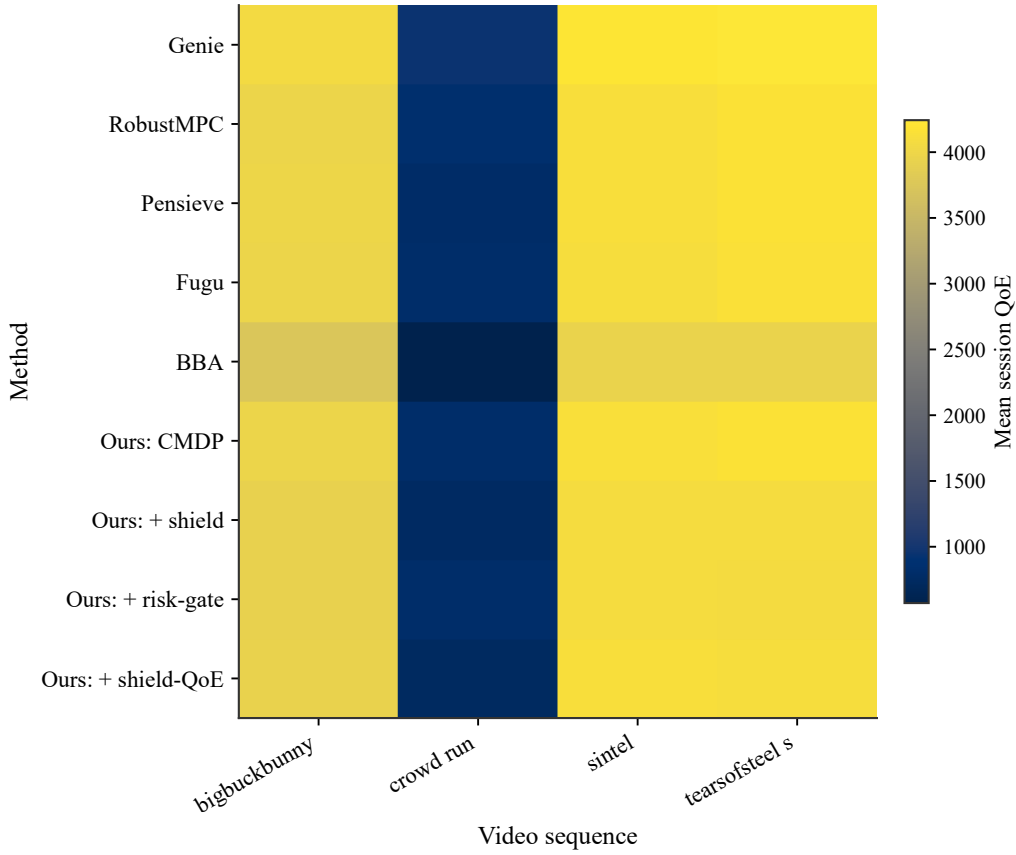


Figure 3: Mean QoE by video and method. QoE is lower across methods for the motion-heavy crowd_run sequence.

Table 7: Paired Wilcoxon signed-rank comparison of VMAF-aware projection ($\sigma_{\text{soft}}=1.0$, $\tau=8$) and legacy projection ($n=80$).

Metric	Mean Δ	Med. Δ	p
QoE (target – legacy)	+42.13	0.00	0.0050
Rebuffer (% pts)	−0.55	0.00	0.0029
VMAF	+0.22	0.00	0.0564

RQ1: Pareto improvement over legacy. Tables 7 and 8 jointly answer RQ1. At $\sigma_{\text{soft}}=1.0$, the mean paired differences are +42.13 for QoE, −0.55 percentage points for rebuffering, and +0.22 VMAF points. The QoE and rebuffering differences are statistically significant, whereas the VMAF difference is not ($p = 0.056$). At $\sigma_{\text{soft}}=0.8$, the corresponding mean differences are +64.82, −0.69 percentage

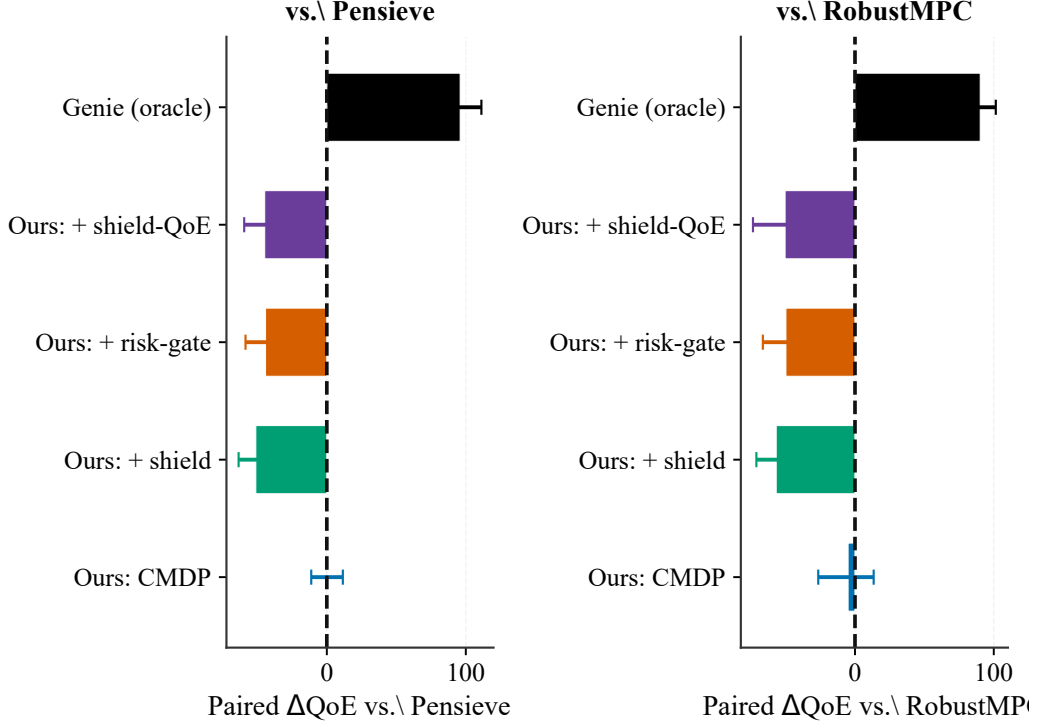


Figure 4: Paired QoE differences relative to Pensieve (left) and RobustMPC (right), with 95% bootstrap confidence intervals within each video and pooled across videos.

points, and +0.32 VMAF points. All three unadjusted tests yield $p < 0.05$. **Engineering interpretation.** This operating point reduces mean rebuffering from 5.67% to 4.98% and increases QoE while preserving perceptual quality: the mean VMAF difference is positive rather than negative. The VMAF effect is below 0.5 points, so the contribution is not a large fidelity increase. Instead, it is the removal of the usual fidelity-for-safety penalty observed with content-blind projection. We describe this finding as empirical quality preservation, not as a formal non-inferiority guarantee, because the study did not prespecify a perceptual margin or perform an equivalence test. Section 5.9.1 evaluates whether the QoE and rebuffering differences persist under synthetic 5G/mmWave stress traces.

Table 8 reports the full sweep. The reported mean operating points for the VMAF-aware family (rows 2–5) empirically dominate the legacy point on QoE, rebuffering, and VMAF. In contrast, threshold-only configurations that vary ρ_{cat} without the VMAF-aware fallback have higher rebuffering ratios than legacy projection (Section 5.7). Within the evaluated grid, enabling the VMAF-aware

fallback accounts for the principal changes in the empirical Pareto set.

Table 8: Mean outcomes for the shield sweep using a fixed policy and paired episodes ($n=80$). VMAF-aware configurations empirically dominate the legacy mean point on the three reported metrics.

Variant	QoE	Rebuf. (%)	VMAF	Interv. (%)
Legacy ($\rho_{\text{cat}}=2$, no VMAF fallback)	2518.87	5.67	65.82	3.23
<i>VMAF-aware</i>				
$\sigma_{\text{soft}}=0.8$, $\tau=8$	2583.70	4.98	66.13	2.79
$\sigma_{\text{soft}}=1.0$, $\tau=8$	2561.00	5.12	66.04	2.71
$\sigma_{\text{soft}}=1.2$, $\tau=8$	2546.86	5.03	65.89	—
$\sigma_{\text{soft}}=1.5$, $\tau=8$	2532.23	5.21	65.86	—
<i>Threshold only (no VMAF fallback)</i>				
$\rho_{\text{cat}}=3$	2519.40	8.94	—	—
$\rho_{\text{cat}}=4$	2530.78	9.96	—	—
$\rho_{\text{cat}}=5$	2557.17	10.49	—	—
<i>Threshold + VMAF fallback</i>				
$\rho_{\text{cat}}=3$	2539.11	8.45	66.30	—
$\rho_{\text{cat}}=4$	2526.21	9.95	66.36	—
<i>Shield off</i>				
Shield off	2563.72	11.04	66.55	0.00

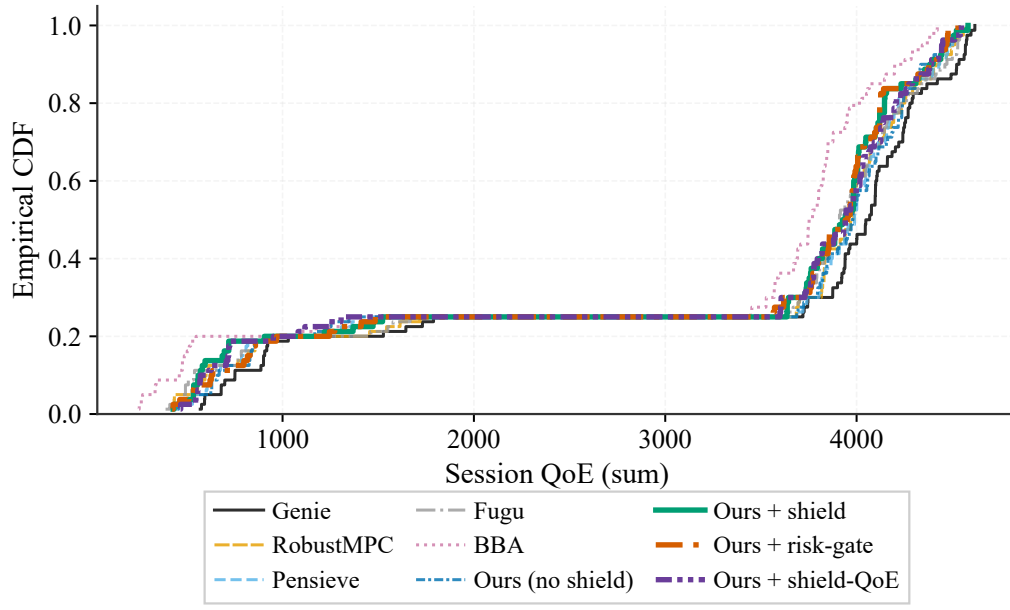
5.5. Tail behavior: empirical CDFs

Figure 5 presents the empirical distributions of session QoE and rebuffering ratio for the distinct learned controllers in Table 4. Figure 6 holds the constrained policy fixed and compares unshielded replay, legacy projection, and VMAF-aware projection with $\sigma_{\text{soft}}=1.0$ and $\tau=8$. Relative to legacy projection, the VMAF-aware QoE curve is shifted to the right, and its rebuffering CDF assigns greater probability to low rebuffering ratios.

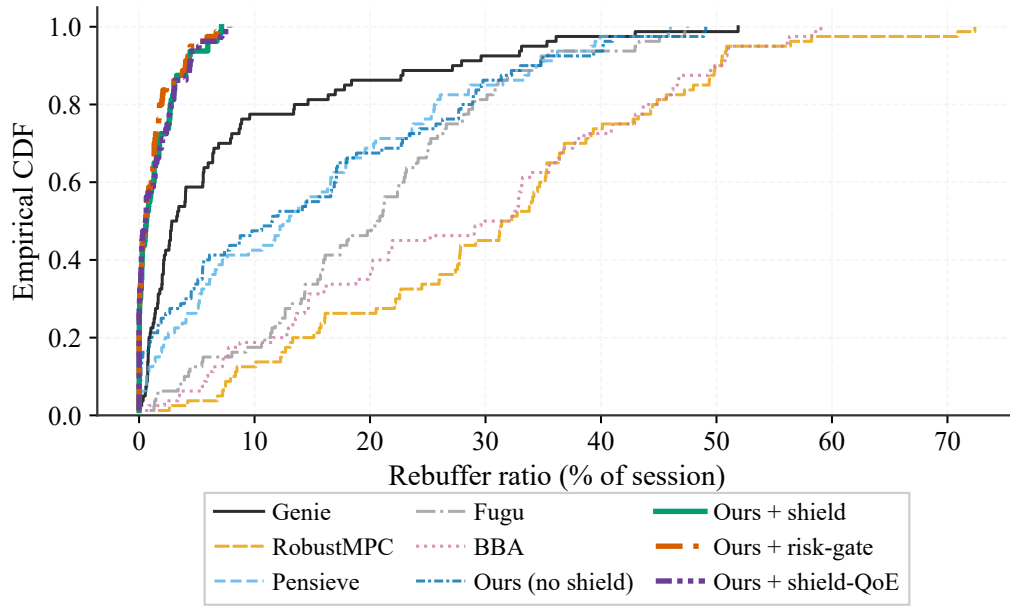
5.6. Pareto dominance visualizations

Figure 7 summarizes the shield sweep in the QoE–rebuffering plane and marks the empirical nondominated set. Figure 8 shows paired episode-level QoE differences relative to legacy projection.

Statistical reporting. Tables 5–6 summarize paired Wilcoxon results; the complete grid is provided as CSV in the reproducibility artifacts. The reported p -values are unadjusted and should be interpreted as exploratory across the full

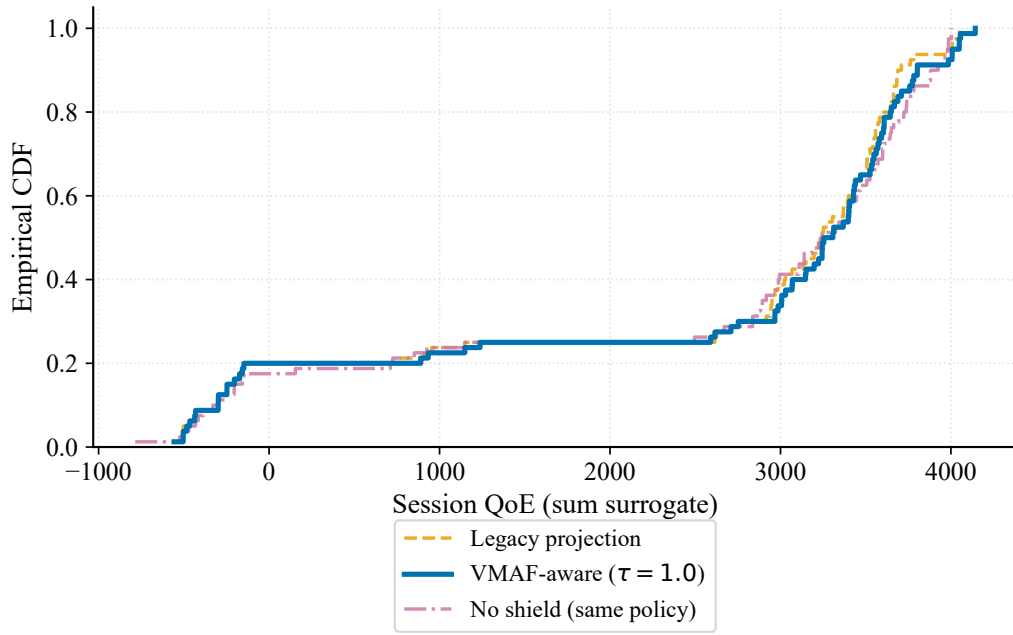


(a) Session QoE

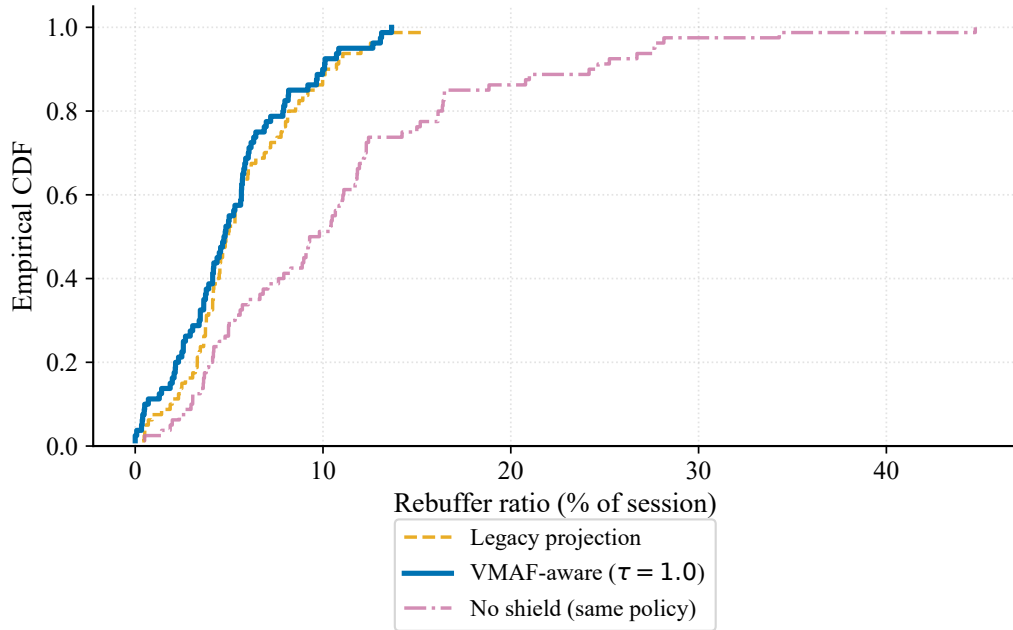


(b) Rebuffering ratio

Figure 5: Empirical CDFs for distinct learned controllers (Table 4). Higher QoE and greater cumulative probability at low rebuffering ratios indicate better performance.

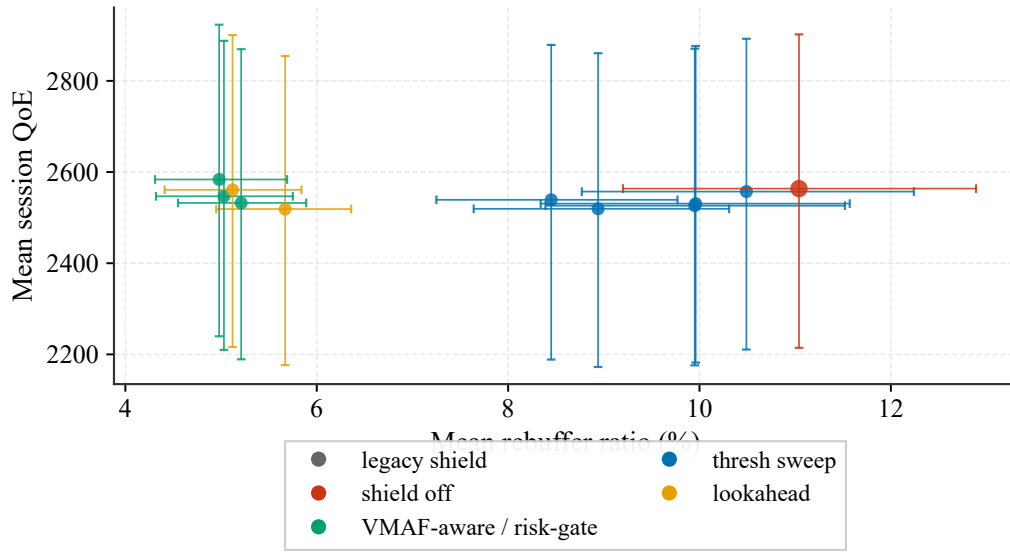


(a) Session QoE

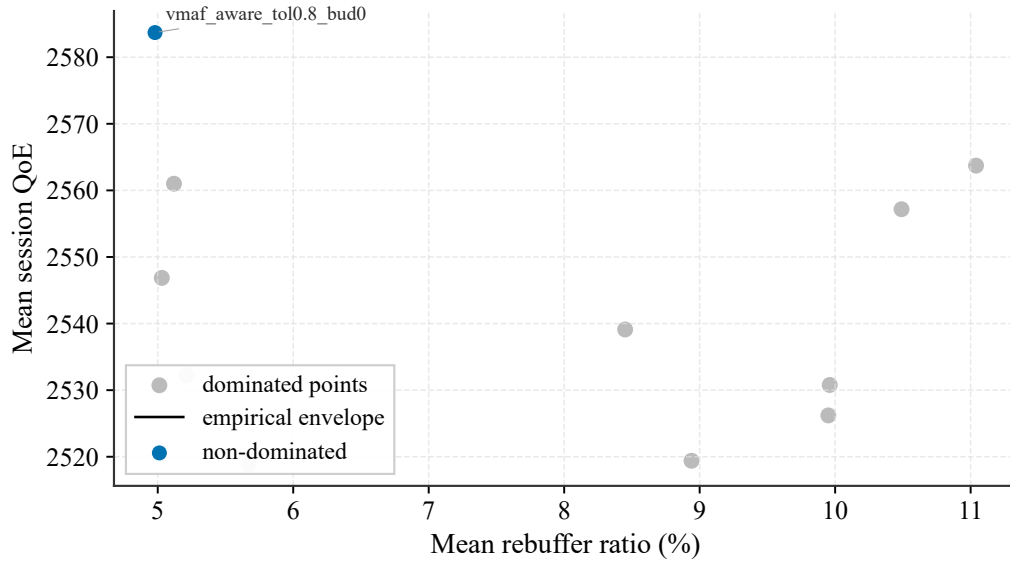


(b) Rebuffering ratio

Figure 6: Paired-replay CDFs ($n=80$). The policy, traces, and seeds are fixed, and only the shield logic varies. VMAF-aware projection assigns greater probability to low rebuffering ratios than legacy projection.



(a) Behavioral groups (95% CIs)



(b) Empirical nondominated set

Figure 7: Shield-sweep Pareto view. (a) Mean QoE and rebuffering ratio for tolerance-defined behavioral groups. (b) Empirical nondominated operating points (blue) and dominated points (gray); connecting lines are visual guides only.

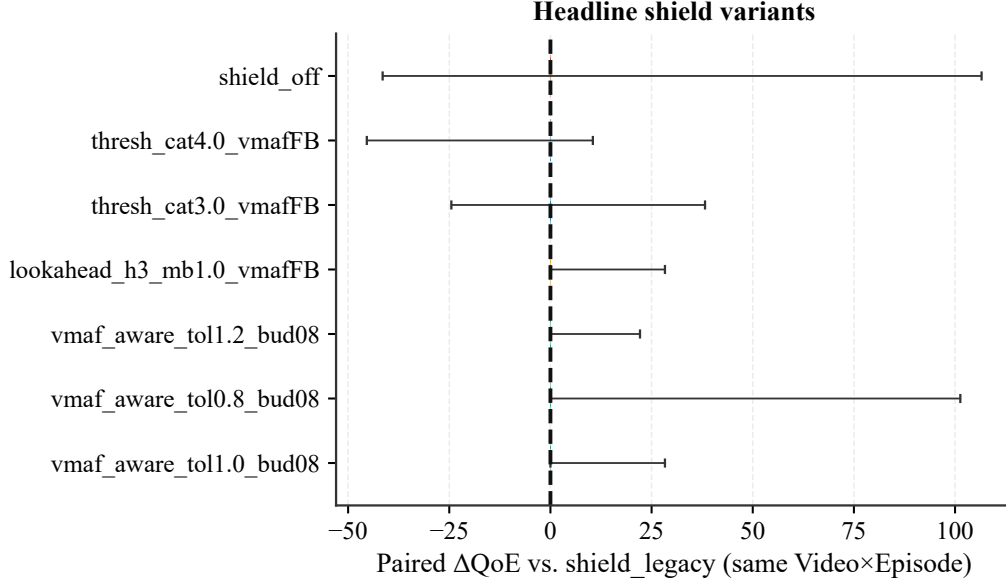


Figure 8: Paired episode-level QoE differences between VMAF-aware and legacy projection ($n=80$); positive values favor VMAF-aware projection.

shield sweep. Because the 80 episodes comprise four videos crossed with 20 seeds, pooled signed-rank tests assume exchangeability across video–seed pairs. The per-video results in Figure 4 and episode-level differences in Figure 8 provide sensitivity views but do not remove this clustering limitation. Under the synthetic stress suite (Sec. 5.9.1), the data support QoE and rebuffering differences but do not establish VMAF equivalence; a prespecified non-inferiority margin and an appropriate equivalence test would be required for that claim [33].

5.7. Shield design-space structure (RQ4)

Sweep design. We instantiate 47 configurations across four main axes. The first two are the catastrophic ratio $\rho_{\text{cat}} \in \{2, 3, 4, 5\}$ and the VMAF-aware fallback with $\sigma_{\text{soft}} \in \{0.8, 1.0, 1.2, 1.5\}$ and $\tau \in \{4, 8, 12\}$. The remaining axes are lookahead pre-gating with $H \in \{2, 3, 5\}$ and associated minimum-buffer thresholds, and the risk-gate settings. Every configuration is evaluated on the same 80 paired episodes.

Behavioral grouping. The 47-point grid yields 11 empirical groups under tolerances of ± 5 QoE points and ± 0.3 rebuffering percentage points. Within the evaluated grid, configurations in the same soft-tolerance band often fall into the same group despite changes in lookahead horizon, minimum-buffer threshold, throughput scaling, or risk gating. After fixing the VMAF-aware fallback and ρ_{cat} ,

the largest observed lookahead-related QoE difference is 1.5 points; risk gating primarily changes intervention frequency.

Effect of the VMAF-aware fallback. At $\rho_{\text{cat}}=3$, enabling the fallback changes mean QoE from 2519.40 to 2539.11 and mean rebuffering from 8.94% to 8.45% (Table 8). This corresponds to +19.71 QoE points and -0.49 rebuffering percentage points. The comparison at $\rho_{\text{cat}}=4$ shows a smaller change. These results support treating the VMAF-aware fallback as a primary architectural setting while retaining σ_{soft} , τ , and ρ_{cat} as configurable parameters. Figure 9 compares the percentage of chunk decisions modified by each shield configuration, quantifying how often the projection intervenes relative to the unshielded baseline.

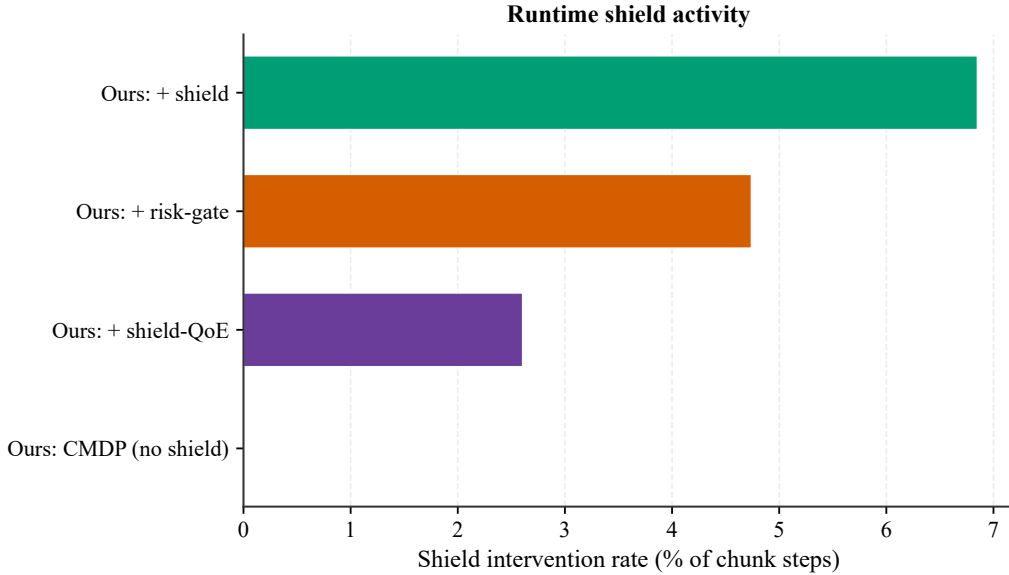


Figure 9: Percentage of chunk decisions modified by each shield configuration; the unshielded baseline has zero interventions.

5.8. Policy-shield co-design (RQ5)

Setup. We evaluate the same shield-trained policy with and without inference-time projection on identical traces and seeds.

Result. Removing the shield increases mean rebuffering from 5.12% to 11.04%. Equivalently, the shielded-minus-unshielded paired difference is -5.92 percentage points with $p = 1.4 \times 10^{-10}$ ($n = 80$). The mean paired QoE difference is +2.4 points and is not statistically significant. Thus, the evaluated policy’s rebuffering performance depends strongly on inference-time projection.

Implication. For this trained policy, deployment should retain the shield and tune σ_{soft} , τ , and ρ_{cat} rather than remove projection. Comparisons involving shield-trained policies should report the shield configuration explicitly.

5.9. Generalization across heterogeneous networks

Broadband versus volatile cellular traces. On the primary FCC/Norway-style suite, the constrained policy has mean QoE comparable to Pensieve, and the shield modifies a small fraction of proposals (Table 3).

Highly volatile traces. On the synthetic 5G/mmWave stress traces, unshielded replay has substantially higher rebuffering, and legacy projection does not eliminate large stall events. The shield intervenes more often when predicted download duration exceeds the available buffer margin, consistent with its risk-based feasibility tests.

5.9.1. Synthetic 5G/mmWave stress replay (OOD)

To evaluate behavior beyond the partitioned broadband traces, we replay the same constrained policy on synthetic 5G/mmWave stress traces containing intermittent LOS/NLOS transitions and outage intervals.

Unshielded comparison. Mean rebuffering is approximately 70.26% without shielding and 21.91% with legacy projection. The difference indicates that the stress suite frequently activates the catastrophic-download logic.

Paired outcomes vs. legacy. At $\sigma_{\text{soft}}=1.0$ and $\tau=8$, the mean paired differences are +34.60 QoE points ($p = 8.1 \times 10^{-7}$) and -2.27 rebuffering percentage points ($p = 2.8 \times 10^{-6}$, $n = 80$). At $\sigma_{\text{soft}}=0.8$ and $\tau=8$, the corresponding differences are +39.85 QoE points ($p = 3.9 \times 10^{-5}$) and -2.38 rebuffering percentage points ($p = 4.4 \times 10^{-6}$). Table 9 lists the corresponding Wilcoxon statistics.

Table 9: Two-sided paired Wilcoxon signed-rank comparisons with legacy projection on synthetic out-of-distribution 5G/mmWave stress traces ($n=80$).

Metric	Mean Δ	Med. Δ	p
$\sigma_{\text{soft}}=1.0$, $\tau=8$			
QoE (target – legacy)	+34.60	+17.28	8.1×10^{-7}
Rebuffer (% pts)	-2.27	0.00	2.8×10^{-6}
VMAF	+0.063	0.00	0.422
$\sigma_{\text{soft}}=0.8$, $\tau=8$			
QoE (target – legacy)	+39.85	+23.12	3.9×10^{-5}
Rebuffer (% pts)	-2.38	-0.48	4.4×10^{-6}
VMAF	+0.014	0.00	0.715

VMAF reporting under stress. The mean paired VMAF differences are fractions of a point, and the two-sided tests do not reject the null ($p = 0.715$ at $\sigma_{\text{soft}}=0.8$; $p = 0.422$ at $\sigma_{\text{soft}}=1.0$). These tests establish neither a VMAF difference nor equivalence. Accordingly, the stress-suite result is limited to statistically significant QoE and rebuffering improvements; a non-inferiority claim would require a prespecified perceptual margin and a dedicated equivalence analysis [33].

5.10. Ablation study (RQ3)

Table 10 examines one-at-a-time component removal from the full CMDP policy. All variants are evaluated without shielding so that the table characterizes policy behavior rather than projection.

Table 10: Mean component-ablation results ($n=80$) on the same traces and videos, without inference-time shielding. Each ablated policy is evaluated after removing one component.

Variant	QoE↑	VMAF↑	Rebuf. (%)↓	Switches↓
Full CMDP (Proposed)	3269.6	71.65	14.73	6.5
– Lagrangian constraints	3260.5	73.58	31.46	1.9
– Manifest lookahead	3224.6	72.01	12.57	12.8
– Lyapunov regularizers	3264.3	70.27	11.63	2.3

– **Lagrangian:** Removing the constraint mechanism increases mean rebuffering from 14.73% to 31.46% while increasing mean VMAF from 71.65 to 73.58. This operating point assigns more weight to fidelity at the expense of rebuffering.

– **Lookahead:** Removing manifest lookahead reduces mean QoE by 45 points and increases the mean number of switches from 6.5 to 12.8, consistent with less stable representation selection.

– **Lyapunov:** Removing the buffer regularizers lowers mean rebuffering from 14.73% to 11.63% but also lowers mean VMAF by 1.38 points.

Interaction-aware interpretation. The ablations reveal a coupled fidelity–stability trade-off rather than three independent additive effects. In particular, removing the Lagrangian constraints eliminates the adaptive penalty on rebuffering violations, allowing the policy to select higher-fidelity but less buffer-feasible representations. Mean VMAF consequently rises from 71.65 to 73.58, but the associated increase in download demand raises mean rebuffering from 14.73% to 31.46%. The stall penalty offsets the fidelity gain, leaving QoE slightly below that of the full policy (3260.5 versus 3269.6). Manifest lookahead and Lyapunov regularization also influence the state trajectories on which the dual variables act: lookahead provides anticipatory chunk-size information, while the regularizers bias exploration

away from unstable buffer regions. Their effects are therefore not expected to sum linearly, because removing one component changes the observations, actions, and constraint violations encountered by the remaining components.

No ablated variant improves all four reported metrics relative to the full policy. Because these are one-at-a-time ablations without uncertainty intervals, they describe component associations and plausible coupling mechanisms rather than identify interaction effects statistically. A factorial ablation with repeated training runs would be required to estimate pairwise or higher-order interactions directly.

Operational complexity. The shield performs deterministic arithmetic and a linear scan over the representation ladder using manifest sizes, buffer state, conservative throughput estimates, and per-representation VMAF values. Its per-decision computational complexity is $O(L)$; runtime latency was not separately benchmarked.

Figure 10 shows how each ablation changes the QoE–rebuffering operating point relative to the full CMDP policy.

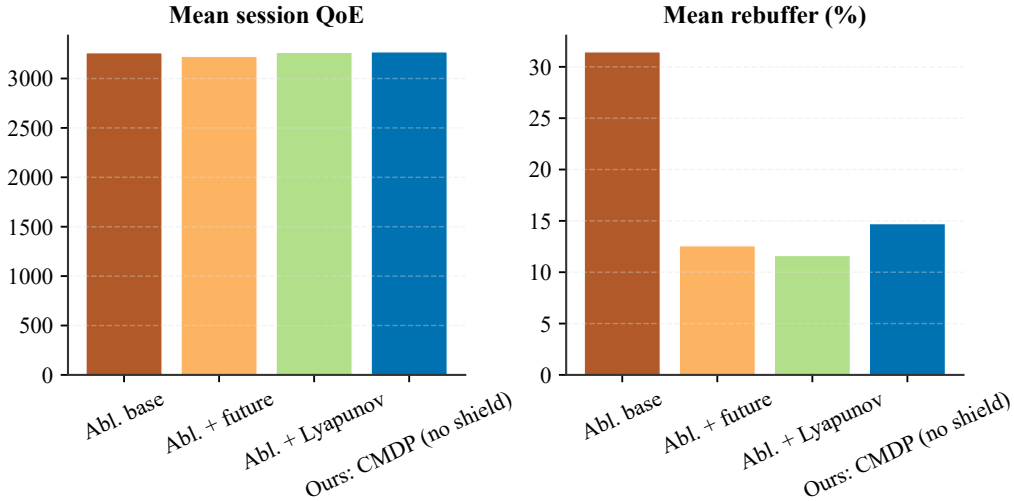


Figure 10: Mean QoE and rebuffering ratio for the full CMDP policy and its one-at-a-time ablations.

5.11. Qualitative case study (time-resolved behavior)

Figure 11 compares buffer and bitrate trajectories on an example broadband episode with and without shielding. During the displayed throughput reductions, the shielded trajectory avoids buffer depletion while selecting the highest representation that satisfies its feasibility test.

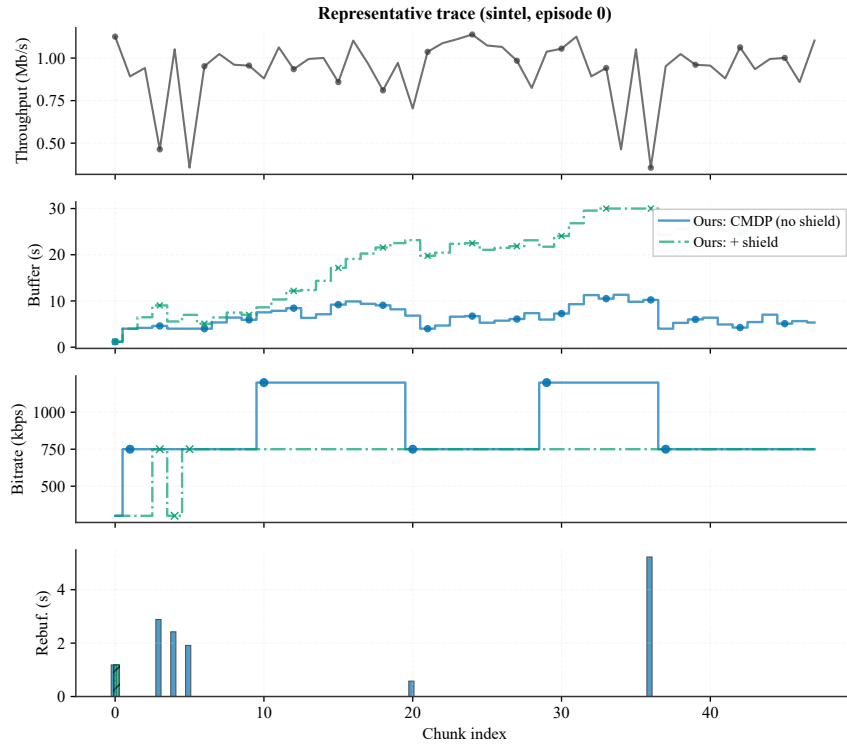


Figure 11: Per-chunk buffer and bitrate trajectories for an example broadband episode. The shielded trajectory (green) avoids buffer depletion during the displayed throughput reductions.

Reproducibility. The reproducibility artifacts include scripts for regenerating all figures.

6. Discussion and limitations

Scope. The proposed shield provides an auditable mechanism for adjusting the QoE–rebuffering trade-off through the VMAF-loss budget τ , soft feasibility tolerance σ_{soft} , and catastrophic-download threshold ρ_{cat} . These parameters can be changed without retraining the policy. The method does not provide a formal worst-case safety guarantee, and the evaluation does not establish dominance over every ABR controller or operating point. Under the reported harness, the mean VMAF-aware operating points empirically dominate legacy projection and have lower mean rebuffering than Pensieve and RobustMPC. The primary broadband result is therefore a quality-preserving safety improvement: the shield reduces stalls and raises QoE without an observed decrease in mean VMAF.

Threats to validity.

1. The evaluation is simulation based and may not capture live cellular dynamics, transport-stack effects, or content delivery network (CDN) cache behavior [3].
2. The test set contains four videos and a finite trace pool. Broader content and trace diversity is needed before deployment-level claims can be made [30].
3. QoE is measured via a VMAF-based surrogate; subjective user studies may rank stall, fidelity, and switching differently [5].
4. The reduction to 11 tolerance-defined behavioral groups is specific to the evaluated policy, ladder, traces, and grouping thresholds. Other ladders, including HDR content, require separate characterization.
5. The reported operating points apply to the joint policy–shield configuration. They may not transfer when the shield is removed or replaced.
6. The synthetic 5G/mmWave suite isolates throughput volatility but does not reproduce a complete network stack. Under severe rebuffering, the observed VMAF differences are small. They establish neither a difference nor equivalence [33].
7. Genie uses non-causal trace knowledge and serves only as an oracle reference.
8. All shield-to-shield comparisons use the same constrained policy. The projection consumes only manifest, buffer, and throughput observations, so applying it to other host controllers is architecturally feasible. However, its performance with Pensieve-, MPC-, or other learned policies has not been evaluated. Such performance should not be inferred from the present results.

9. The paired episodes share four videos and a common trace pool, so pooled tests assume exchangeability across video–seed pairs. Per-video breakdowns provide a sensitivity analysis but do not eliminate this dependence.

Future work. First, live or packet-level emulation and field-collected 5G/mmWave traces are needed to evaluate transport and deployment effects [3]. Second, applying the projection to heterogeneous host policies, including Pensieve, MPC, and quality-aware learners such as Comyco [25], would separate its standalone contribution from policy–shield co-design. Third, future controllers could incorporate per-trace tail metrics, risk-sensitive objectives such as CVaR [34], uncertainty-aware throughput prediction, and telemetry-driven adaptation of τ .

7. Conclusion

We introduced a deterministic VMAF-aware runtime shield for a constrained DRL-based ABR controller. The shield uses manifest VMAF values, conservative download-time estimates, and buffer-feasibility tests to select an emergency downgrade rather than relying only on representation-index order.

On 80 paired broadband episodes, the configuration with $\sigma_{\text{soft}}=0.8$ and $\tau=8$ reduces mean rebuffering from 5.67% to 4.98% and increases mean QoE by approximately 2.6% relative to legacy projection. It achieves these gains without an observed reduction in perceptual quality: mean VMAF increases slightly, and all three unadjusted paired tests yield $p < 0.05$. This quality-preserving reduction in stalls is the principal engineering benefit of the VMAF-aware projection. Under synthetic 5G/mmWave stress traces, the QoE and rebuffering improvements remain statistically significant, whereas VMAF is not statistically distinguishable from legacy projection. The shield sweep further shows that enabling the VMAF-aware fallback is associated with the principal changes in the empirical Pareto set.

For the evaluated shield-trained policy, removing inference-time projection increases mean rebuffering from 5.12% to 11.04%. Deployment should therefore retain the shield as part of the joint policy–shield configuration. Future work should test this projection with other host policies and under packet-level or live-network conditions.

Data availability

The reproducibility artifacts include the code, evaluation scripts, and exported CSV files underlying Tables 3–10. The deterministic shield implementations and their parameters are documented with the code.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Leili Farzinvash:** Supervision, Writing – review & editing. **Pedram Salehpour:** Conceptualization, Supervision, Project administration, Writing – review & editing.

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